

Problem Set 05 Discussion

30 September 2025

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Data Cleaning

- Removing NA values:
 - colSums, etc.

```
gasch1k_trimmed <- gasch1k |>
  select(where(~ sum(is.na(.)) <= 10))
```

```
data_clean <- data[, colSums(is.na(data)) <= 10]
```

- Matching minute values
 - grepl

```
gasch1k_min <- gasch1k_trimmed |>
  filter(grepl("min|minutes", experimental_condition))
```

- Advanced – string wrangling
 - stringr

```
gasch.wide <- long.gasch |>
  mutate(minute_value = stringr::str_extract(experiment.conditions, "(?<=\\.\\.\\d+(?=\\.\\. (min|minutes))"))
head(gasch.wide)
```

Yeast Joining

- Sorry for the confusion!
- Intended dataset to join - go_slim_mapping.tab
- Filter on nucleus genes
 - Will give you 237 or 1168
- Filter rRNA genes
 - Will give you 0

```
left_join(spellman, GO_wide, by = join_by(gene_ID)) |>  
  filter(cellular_component == "nucleus") |>  
  filter(alpha7 < 0) |>  
  nrow()
```

```
left_join(spellman, GO_wide, by = join_by(gene_ID)) |>  
  filter(molecular_function == "RNA binding") |>  
  filter(alpha7 == min(alpha7, na.rm = TRUE))
```